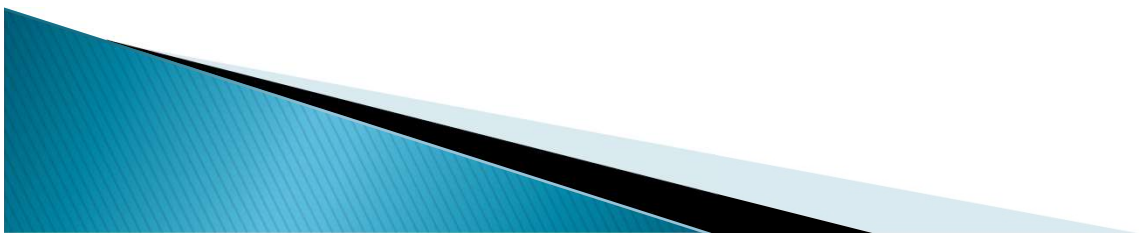
Introduction to lymphatic system



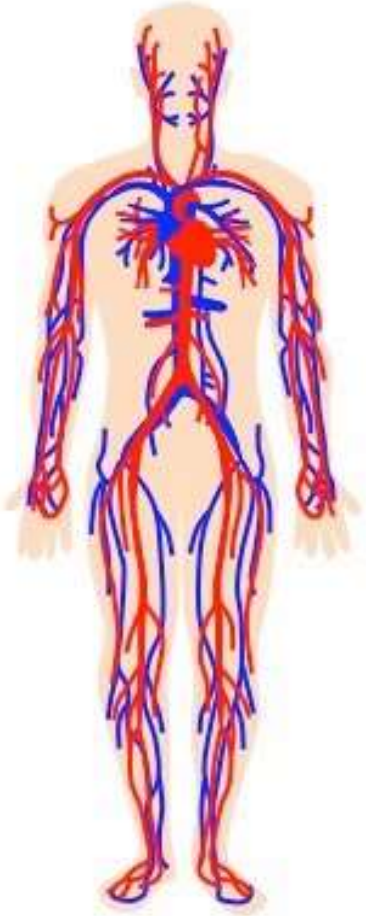
What is lymphatic system?

- Lymphatic system is essentially a drainage system that carries excess tissue fluid back to the circulating blood.
- Most of the tissue fluid formed at the arterial end of capillaries is absorbed back into the blood by venous system, the remaining tissue fluid (10-20%) is absorbed by the lymphatics.
- It provides an important immune mechanism for the body.

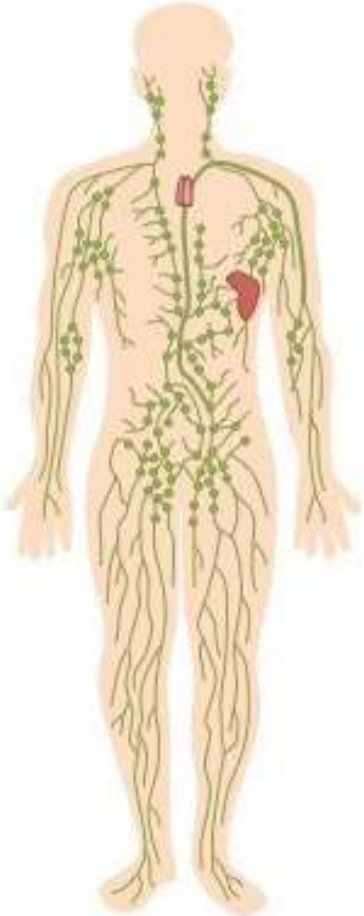
- The larger particles(proteins and particulate matter) can be removed from tissue fluid only by the lymphatics. Therefore the lymphatic system may be regarded as “Drainage system of coarse type”.

- Where as venous system is regarded as, “Drainage system of fine type”.

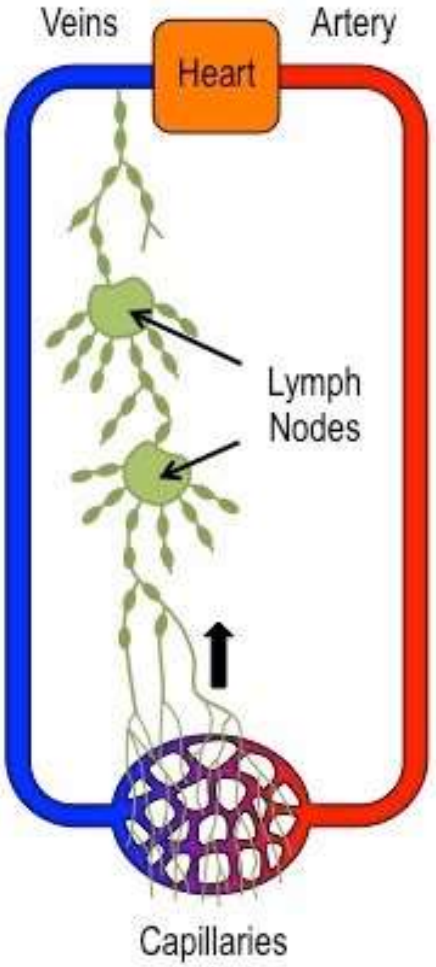
Circulatory System



Lymphatic System



Inter-relationship between systems



LYMPH FLOW FROM TISSUE

TISSUE FLUID



LYMPH



LYMPHATIC CAPILLARY



AFFERENT LYMPHATIC VESSEL



LYMPHNODE



EFFERENT LYMPHATIC VESSEL



LYMPHATIC TRUNK



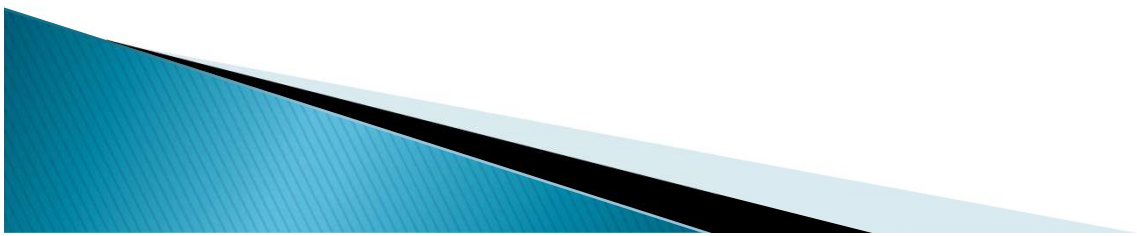
COLLECTING DUCT



BRACHIOCEPHALIC VEIN

Components of lymphatic system

1. Lymph
2. Lymph vessels
3. Primary or Central lymphoid tissues/organs
4. Secondary or Peripheral lymphoid tissues/organs
5. Circulating lymphocytes



1.Lymph

- Tissue fluid flowing in the lymphatics is called lymph.
- Lymph from most tissues is clear and colourless.
- Lymph from small intestine is milky-white due to absorption of fat and is called as “chyle”.

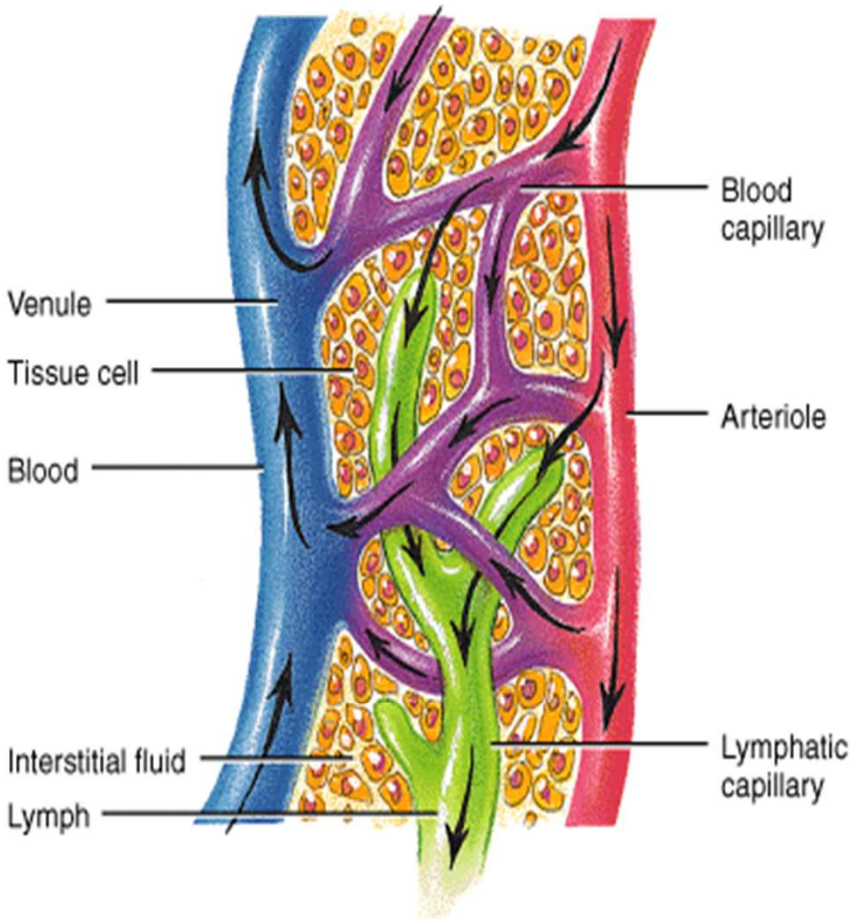
2.Lymph vessels

- Composed of:

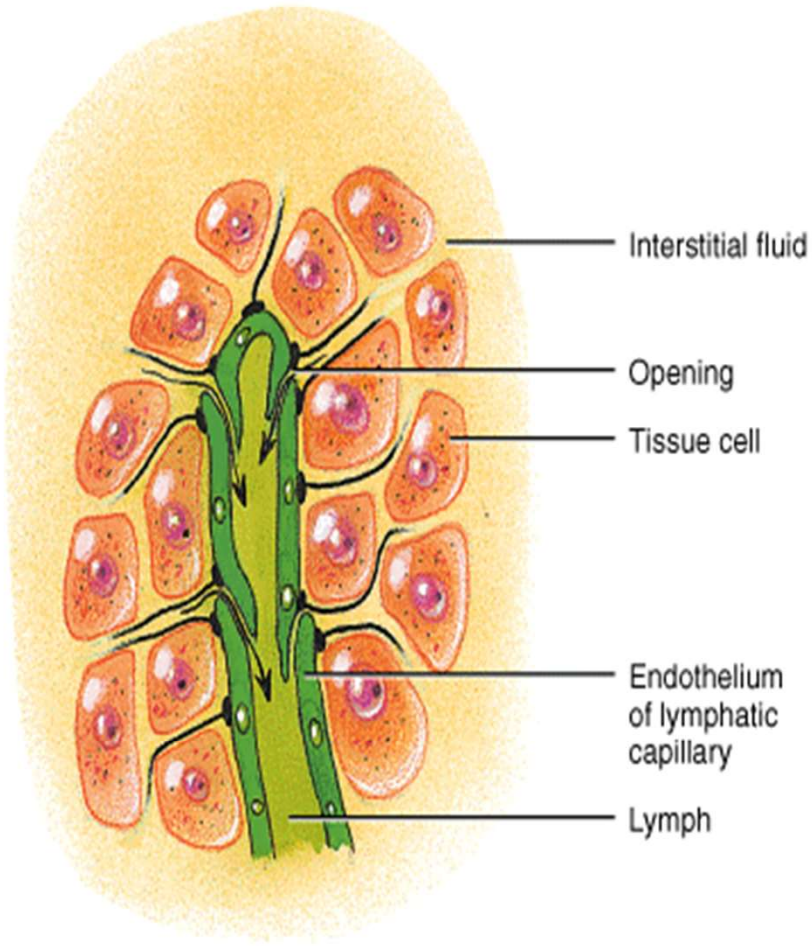
A. Lymphatic capillaries
B. Lymphatic collecting vessels (superficial and deep)
C. Lymphatic trunks and lymphatic ducts

A. Lymph capillaries

- Lymph capillaries begin blindly in the tissue spaces and form intricate networks.
- Their endothelial wall is permeable to substances of much greater molecular size.
- these are absent in bone marrow, cartilage, nail, hair, cornea, splenic pulp, CNS (brain, spinal cord).
- Lymph capillaries in the villi of small intestine are called lacteals, where they absorb emulsified fat.
- Consists valves which prevents the back flow of lymph.
- Nature of lymphatic capillaries allow WBCs, pathogens and cancer cells to enter easily.



(a) Relationship of lymphatic capillaries to tissue cells and blood capillaries



(b) Details of a lymphatic capillary

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Comparison of lymph and blood capillaries

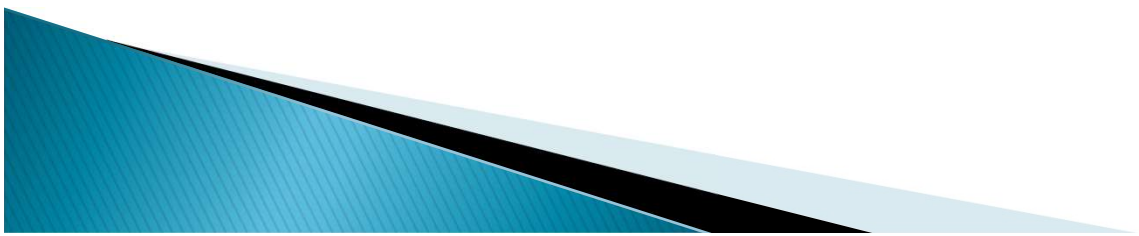
Lymph capillaries	Blood capillaries
1. Colourless, difficult to observe.	1. Reddish, easy to observe.
2. Blind (closed at the tip).	2. Joined to arterioles at one end and to venules at another end.
3. Wider than blood capillaries.	3. Narrower than lymph capillaries.
4. Wall consists of thin endothelium and poorly developed basement membrane.	4. Wall consists of normal endothelium and basement membrane.
5. Contain colourless lymph.	5. Contain red blood.
6. Have relatively low pressure.	6. Have relatively high pressure.
7. Absorb tissue fluid from inter-cellular spaces.	7. Add tissue fluid to inter-cellular spaces.

B. Lymph vessels/lymphatics

The lymph capillaries join to form lymphatics, which are:

- SUPERFICIAL LYMPHATICS: these accompany veins.
- DEEP LYMPHATICS: these accompany arteries.

Collection of many lymph vessels is called as “LYMPHATIC TRUNK”. It drains into the lymphatic Ducts.



C. Lymphatic Ducts

The filtered lymph passes through lymphatics (lymph vessels, lymphnodes, lymphatic trunks) and is eventually collected into two large Lymphatic Ducts.

1. THORACIC DUCT: drains both lower limbs, abdomen, left half of thorax, left half of head and neck and left upper limb.
   - Enters at the junction of left internal jugular and left subclavian vein ⟹ left Brachiocephalic vein

2. RIGHT LYMPHATIC DUCT: drains right half of thorax, right half of head and neck, right upper limb.
   - Enters at the junction of right internal jugular and right subclavian vein ⟹ right Brachiocephlic vein

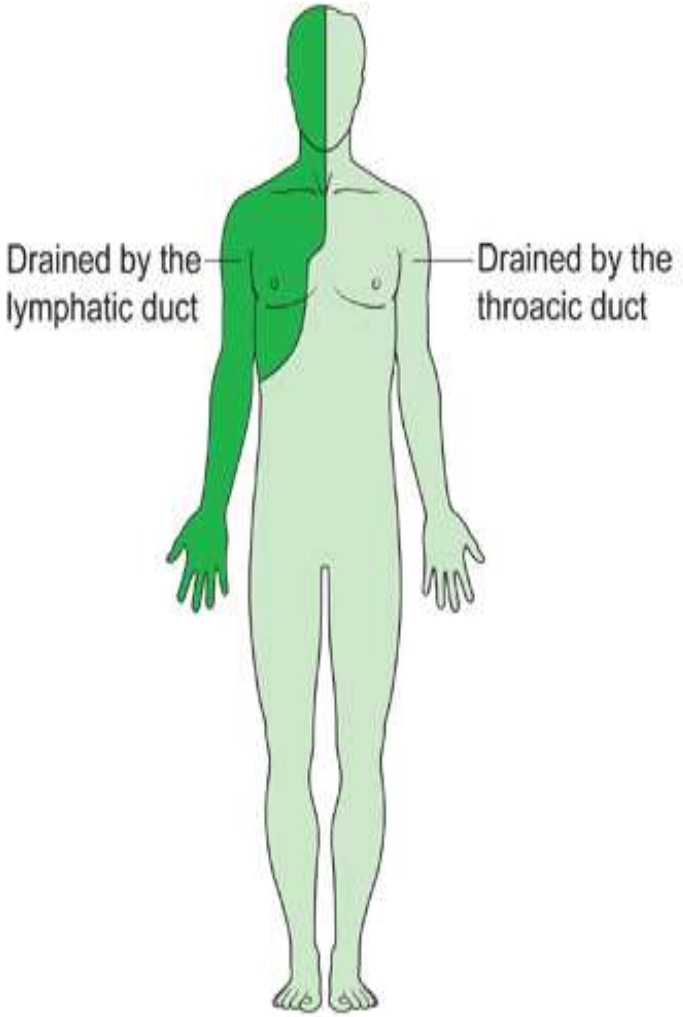
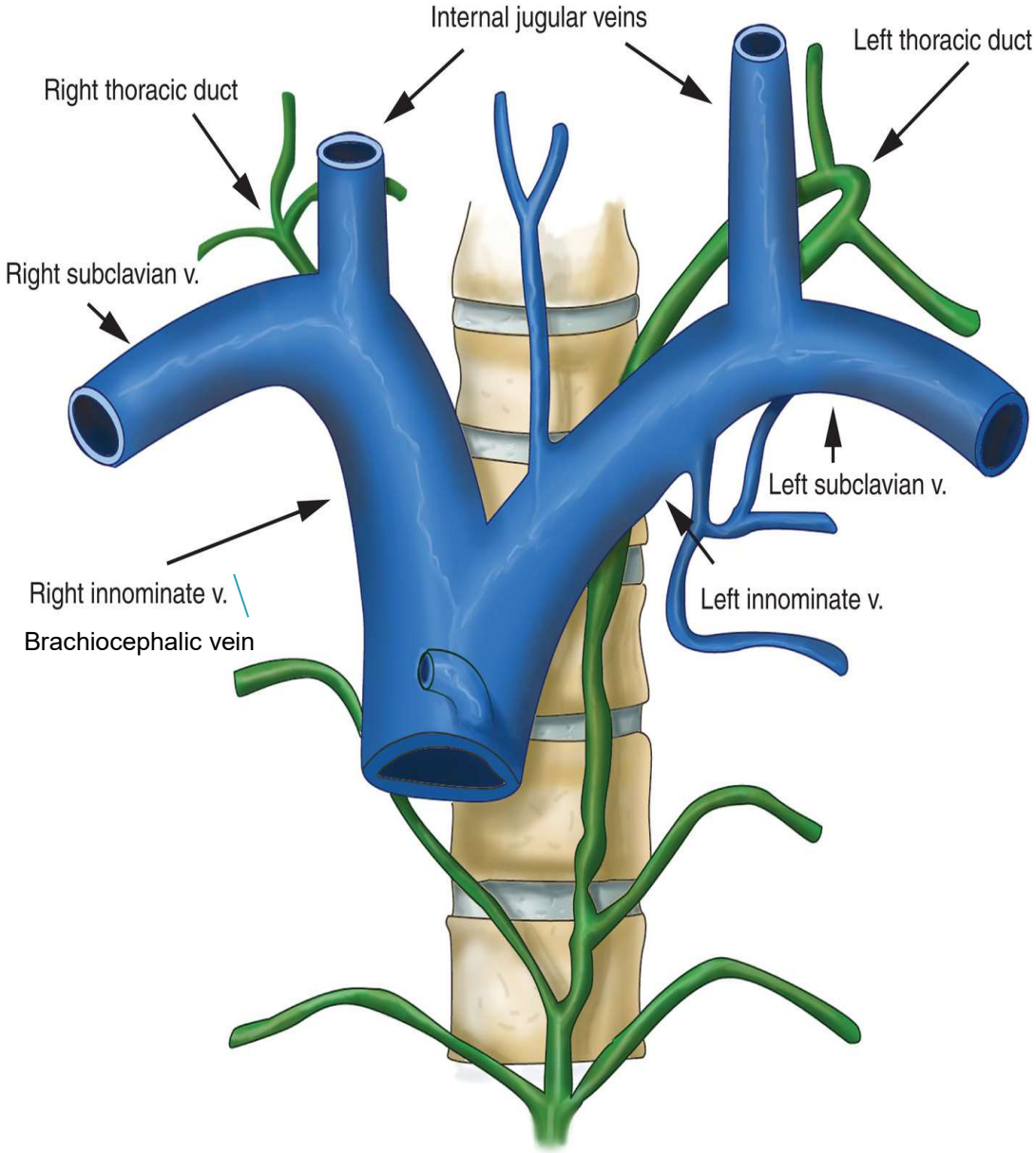


Diagram showing the parts of the body the lymphatic and thoracic ducts drain
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3. Primary/Central lymphoid tissues

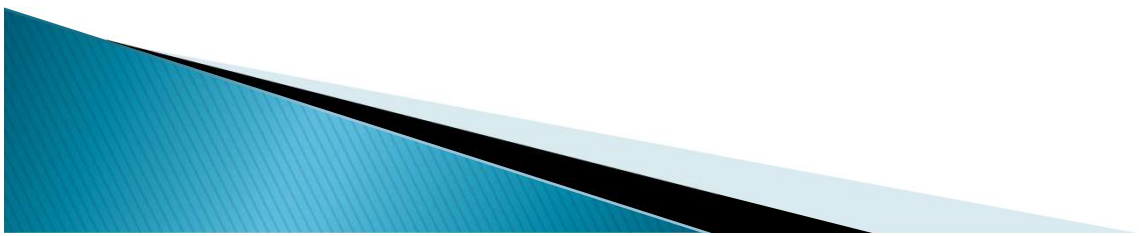
It comprise of:

1.BONE MARROW:

- All pluripotent stem cells(lymphoid and myeloid progenitor cells) are initially produced by bone marrow except early fetal life where these are produced by liver and spleen.
- Bone marrow helps differentiation of B-lymphocytes.

2.THYMUS: it helps in the differentiation of T-lymphocytes.

The lymphoid stem cells undergo differentiation in the central lymphoid tissues and then are supplied to secondary lymphoid organs.



4. Secondary/Peripheral lymphoid organs

It comprises of:

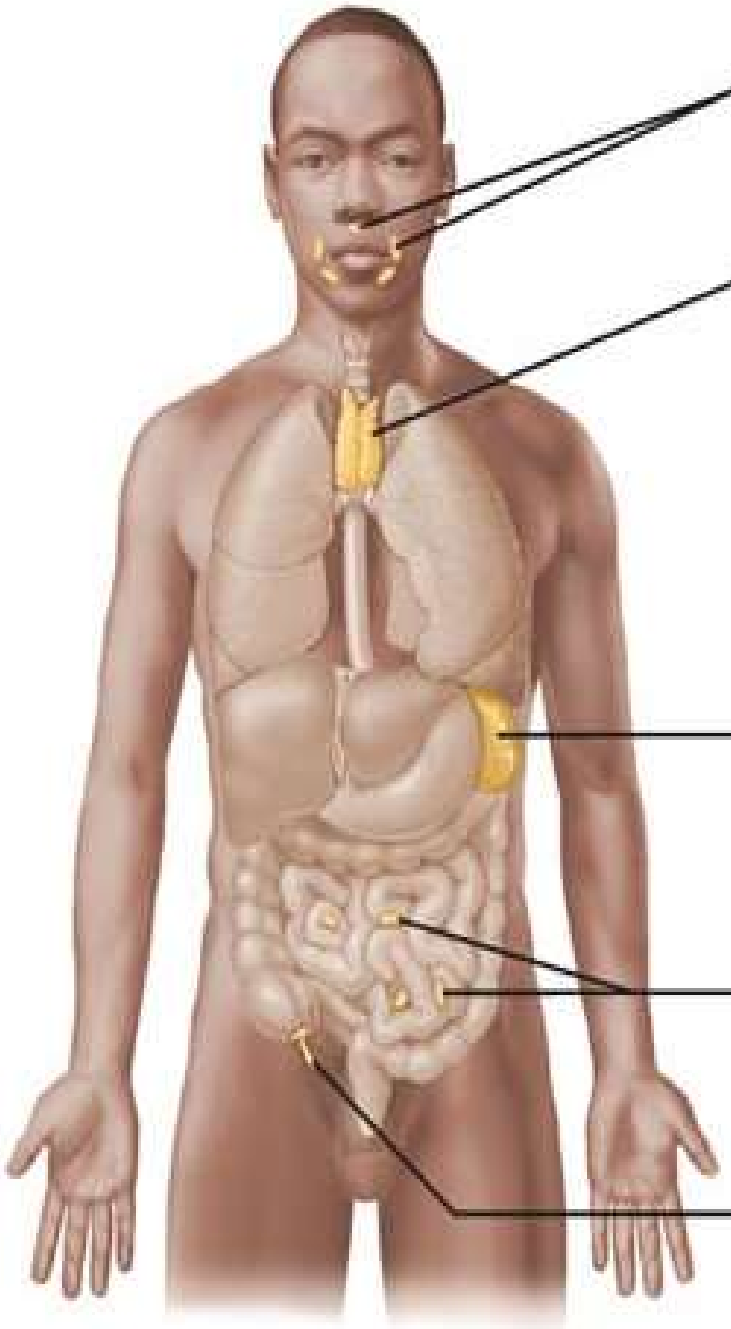
1. Lymphnodes
2. Spleen
3. Tonsils
4. Epithilio-lymphoid tissues

>>>lymphoid nodules developed in the alimentary tract (peyer's patches in ileum, appendix) and respiratory tract

- The progenies of B and T- lymphocytes reach these organs and proliferate and mature to become competent defensive elements of the immune system. –the mature lymphocytes join the "circulating pool of lymphocytes".

5. Circulating pool of lymphocytes

- The pool contains the mature progenies of B- and T-lymphocytes.
- These mature lymphocytes may be called upon during antigenic emergencies.



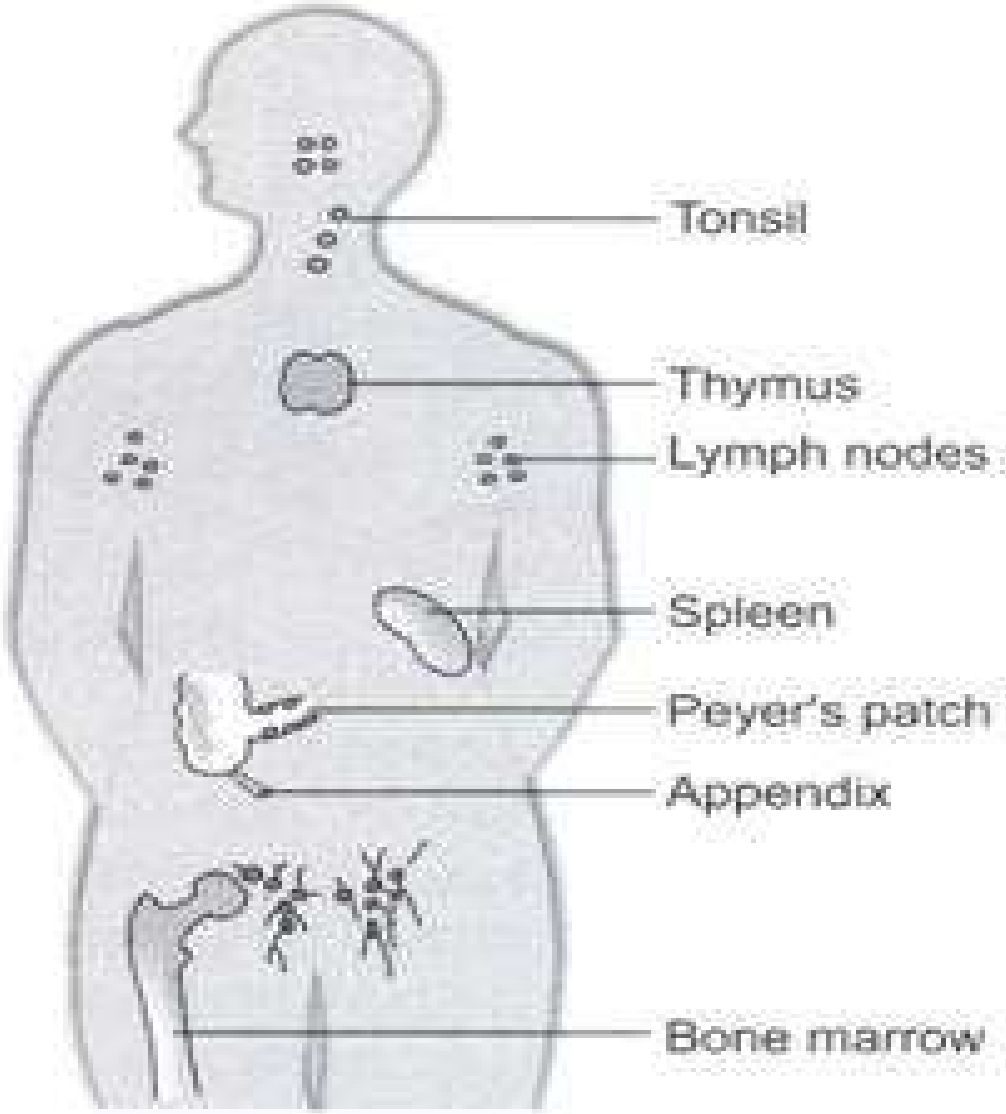
Tonsils (in pharyngeal region)

Thymus (in thorax; most active during youth; almost entirely replaced with fibrous and fatty tissue by age 40)

Spleen (curves around left side of stomach; largest lymphoid organ; removes antigens and old/defective blood cells from blood; stores platelets)

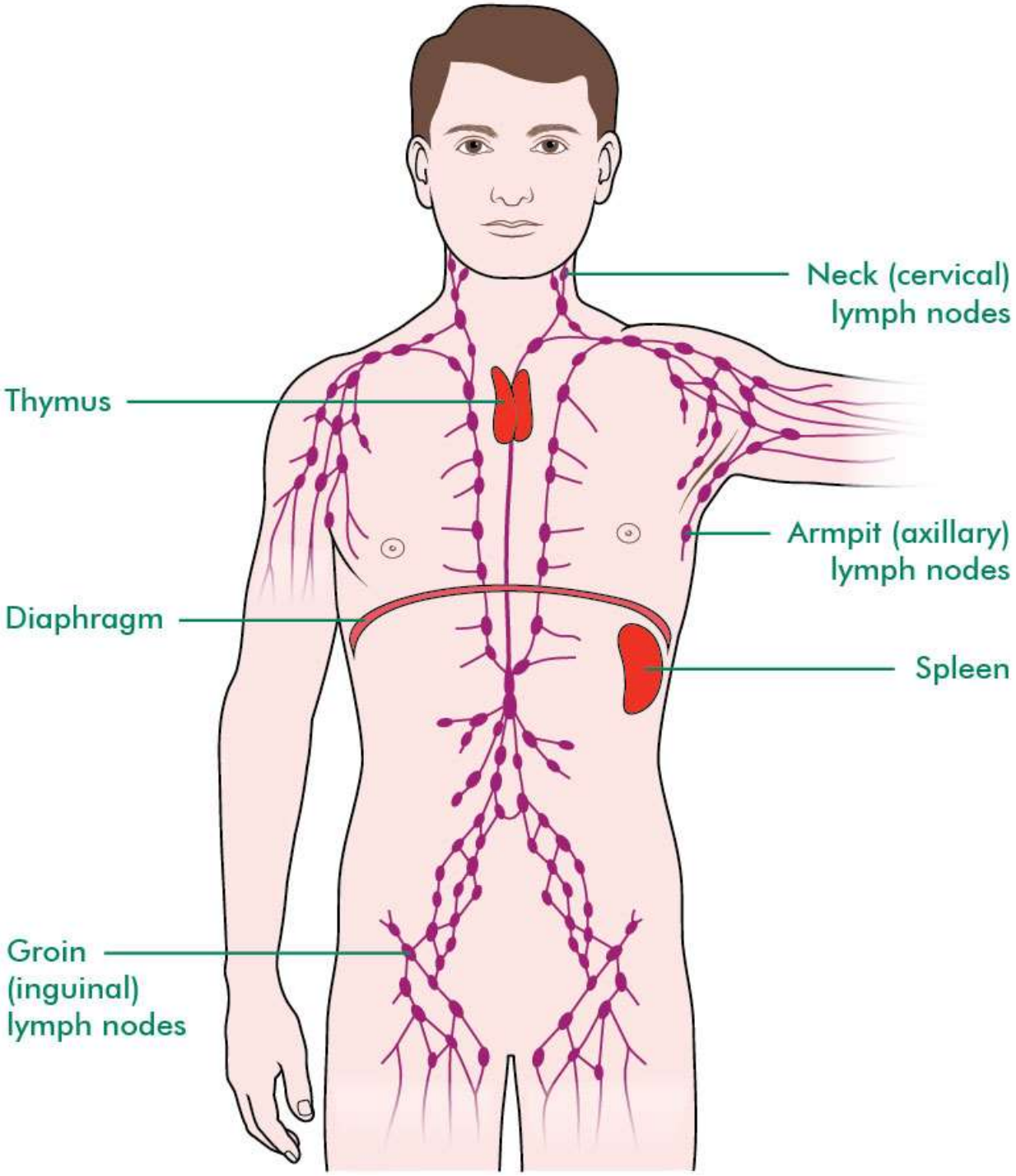
Aggregated lymphoid nodule (in intestine)

Appendix


Tonsil
Thymus
Lymph nodes
Spleen
Peyer's patch
Appendix
Bone marrow

Lymphnode

- It is the principal filtering organ of lymphatic system
- It is a small nodule of lymphoid tissue found in the course of smaller lymphatics.
- Lymph passes through one or more lymph-nodes before reaching the larger lymph trunks.
- They usually occur in groups eg: cervical, axillary, inguinal, messenteric, mediastenal etc.
- Superficial LN are arranged along the veins and deep LN along the ateries.

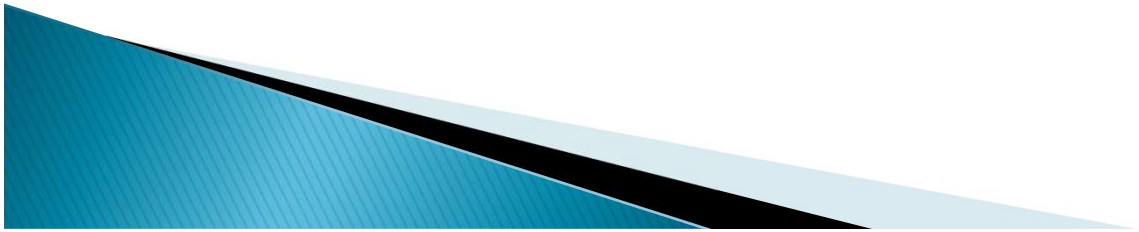

Neck (cervical) lymph nodes
Thymus
Armpit (axillary) lymph nodes
Diaphragm
Spleen
Groin (inguinal) lymph nodes

Structure of lymphnode

The nodes are oval or reniform in shape and is made up of following parts:

1. Fibrous and reticular framework:

- The lymphnode is covered by a fibrous
- capsule.
- From the deep surface of capsule a number of trabeculae extend radially into the interior of the node and divide the node into compartments and then trabeculae are continuous with the fine reticulum which forms the supporting framework for the lymphoid tissue.

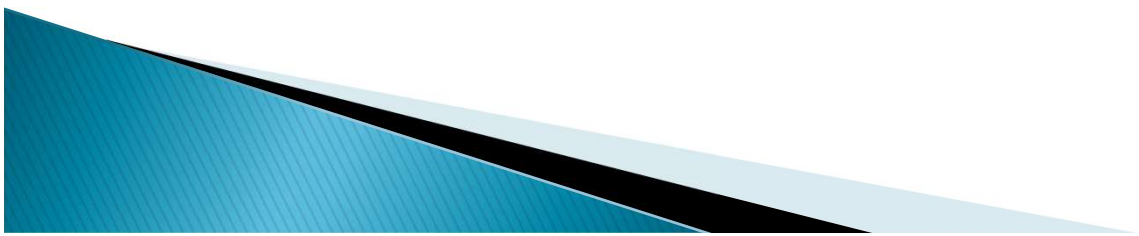


2. Lymphatic channels:

- The sub-capsular sinus lies beneath the
- capsule and surrounds the node except at the hilum.
- Many afferent lymphatics of the node open into the subcapsular sinus.

3. Cortex:

- It is the outer part of the lymphnode situated beneath the subcapsular sinus, being absent at the hilum.
- It is made up of lymphatic follicles and is traversed by fibrous trabeculae.
- Contains lymphocyte, macrophages, and ocasional plasma cells.

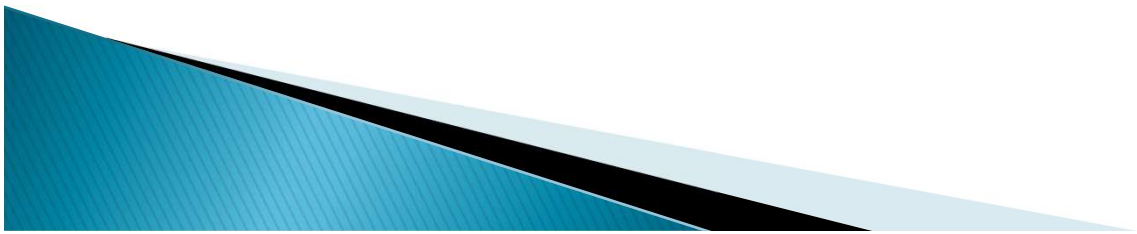


4. Medulla:

- It is the central part of the lymphnode, containing loosely packed lymphocytes, plasma cells and macrophages.

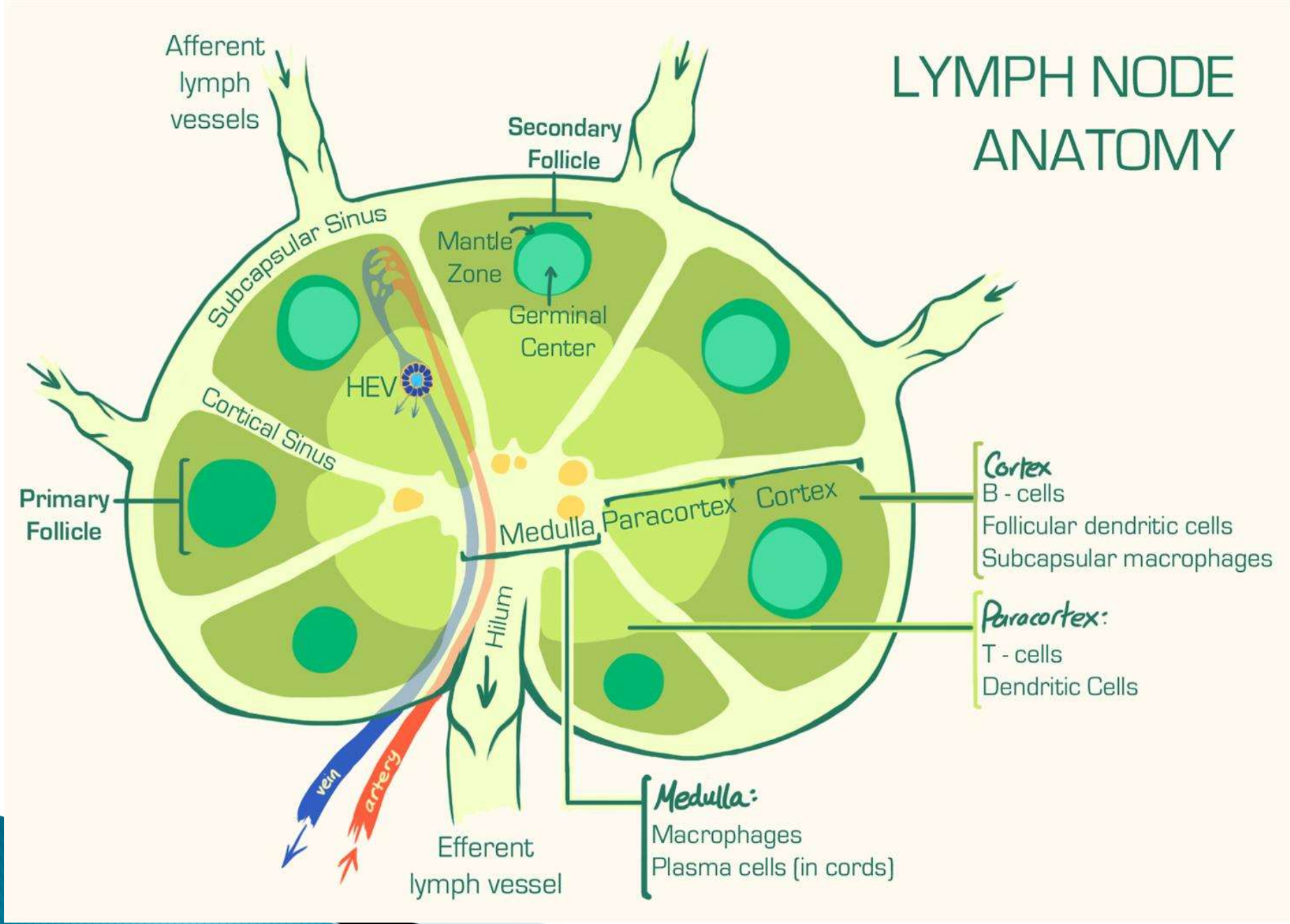
5. Hilum:

- Each lymphnode has a slight depression on one side, called hilum.
- At the hilum, artery enters the node and the vein with efferent lymphatic come out of it.



6. Blood channels:

- The artery enters at the hilum and divides into straight branches which run in the trabeculae.
- In the cortex arteries further divides to form arcades of arterioles and capillaries with many anastomosing loops.
- The capillaries give rise to venules and veins, which run back to the hilum.


LYMPH NODE ANATOMY
Afferent lymph vessels
Secondary Follicle
Subcapsular Sinus
Mantle Zone
Germinal Center
HEV
Cortical Sinus
Primary Follicle
Medulla
Paracortex
Cortex
Hilum
vein
artery
Efferent lymph vessel
Cortex
B - cells
Follicular dendritic cells
Subcapsular macrophages
Paracortex:
T - cells
Dendritic Cells
Medulla:
Macrophages
Plasma cells (in cords)

Path of Lymph Flow through a Lymph Node

6. **Efferent lymphatics** (*efferens*, to bring out) leave the lymph node at the hilum. These vessels collect lymph from the medullary sinus and carry it toward the venous circulation.

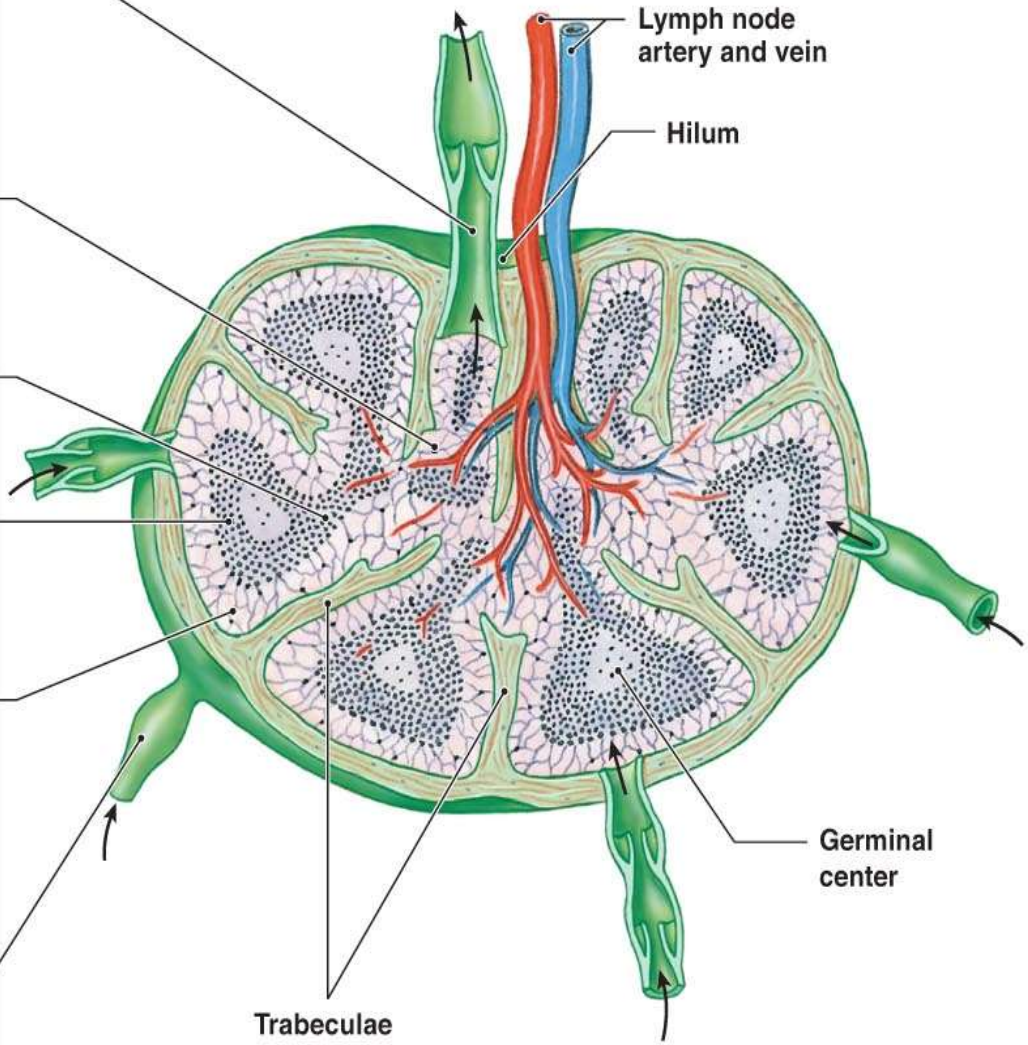
5. Lymph continues into the **medullary sinus** at the core of the lymph node. This region contains B cells and plasma cells.

4. Lymph then flows through lymph sinuses in the **deep cortex**, which is dominated by T cells.

3. Lymph next flows into the **outer cortex**, which contains B cells within germinal centers that resemble those of lymphoid nodules.

2. The afferent vessels deliver lymph to the subcapsular space, a meshwork of reticular fibers, macrophages, and dendritic cells. **Dendritic cells** are involved in the initiation of the immune response.

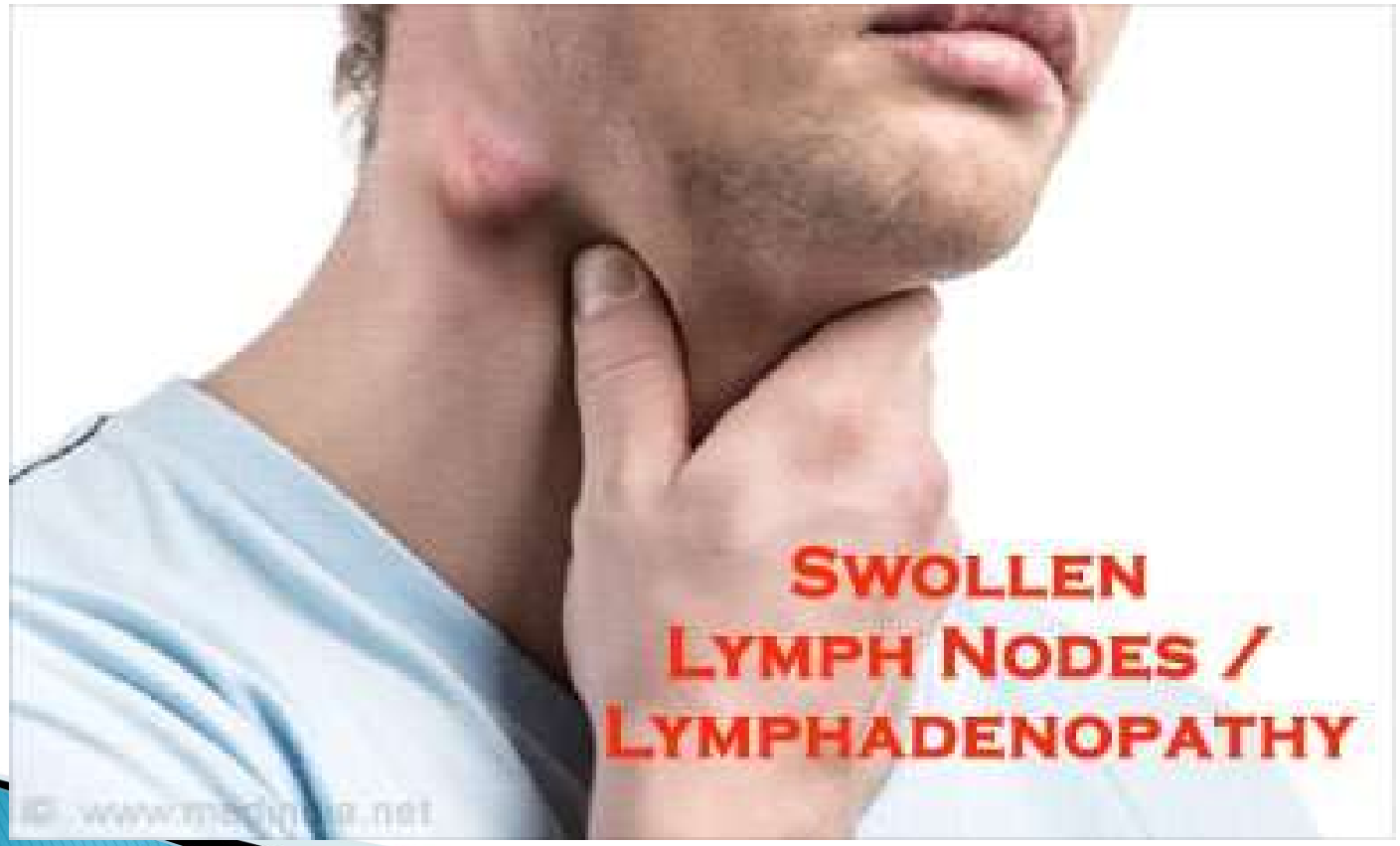
Start 1. **Afferent lymphatics** (*afferens*, to bring to) carry lymph to the lymph node from peripheral tissues. The afferent lymphatics penetrate the capsule of the lymph node on the side opposite the hilum.

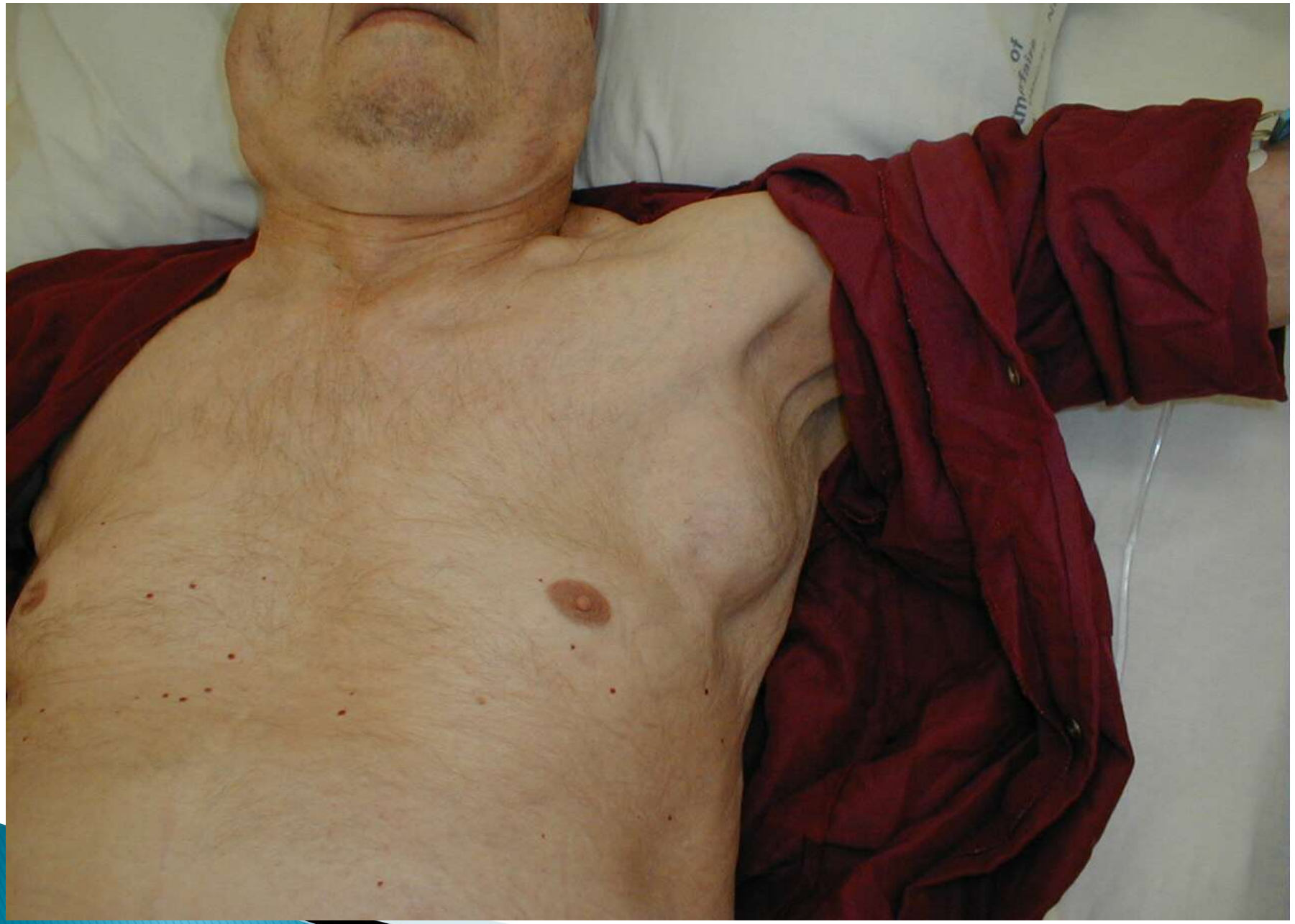


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Lymphadenitis

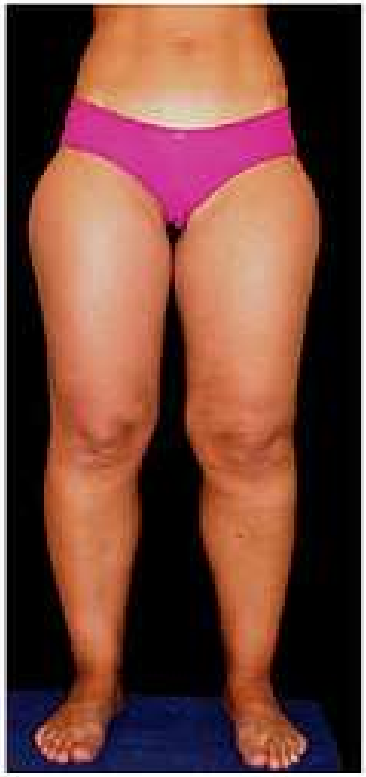
- Enlargement of one or more lymph-nodes due to infection or inflammation.





Elephantiasis/ flariasis

Enlargement of lower limb and scrotum due to obstruction of lymphatic vessels by pasarasitic worms known as filarial worms.



Stage 1



Stage 2



Stage 3



THANK YOU

